① Inter Process Communication.

- within
- ?
- Info share.

Process: 1 → Program.

↳ Set of Ins.

↳ run state Inst.

There may be multiple states.

↳ 4 - State Model

↳ 5 ——

↳ 6 ——

→ Instruction → current Status

↓

Process Control Block

PCB

PCB: All information related to the process

(process ID, states, registers, etc.)

2

Network: More than one device connected via link.

link ⟶ network.
↓
LAN, MAN, WAN.

→ collection of interconnected devices

→ network allows to share the information.

Information sharing
↓

IPC
Inter process communication
(uni-device)

RPC
Remote process communication
(network connected device)

(5)

IPC :-

Sender is always a Sender
Receiver ——— Receiver

Methods i → Pipe

ii → Queue

iii → Shared Memory

iv → Semaphore
↳ control of concurrent Access.

v → Socket (Network virtual design).

i → // Socket creation
socket ( )

ii → bind ( ) // IP add

iii → listen ( )

iv → Accept ( )

v → Send ( ). or
Recv ( )

vi → close ( )

(4)

RPC

one device ⟷ Second device

H/w

Req → RPC mechanism

i → Synchronous RPC,
↳ client - server model.
client wait till server's response.

ii → Asynchronous RPC,
↳ client → req → server
parallelly:
optional

Server replay back,
↳ client will process the server response.

H/W
→ OSI Model → iso certified
→ TCP/IP Model → IETP certified

OSI → 7 layer. ⇒ protocol indep.

TCP/IP → 4-layer ⇒ protocol dep.

(5) OSI

i - Physical layer
↳ Sensors, H/W

Protocols. ⇒ Ethernet, 802.11 (wifi)

ii - Data link layer
→ Node to node data link
→ Data error detection

Protocols ⇒ MAC, ethernet

iii → Network layer.
→ IP address assigning

Protocols ⇒ IP, ICMP

iv → Transport layer
→ Data transfer (info shares)

Protocols ⇒ ~~[illegible]~~, U.D.P, TCP/IP.

v → Session layer
→ Sessions will be observed in this layer

Protocols → RPC

vi → Presentation layer
→ Data Encryption & Data compression

protocols ⇒ SSL / TLS → JPEG.

(6) vii → Application layer

→ user end application

HTTP, SMTP, FTP.

TCP/IP layer :

① → link layer (physical + Data link)

② → Network layer

③ → Transport layer

④ → Application (Session + Presentation + Application)